



Systems and Control integration project (5SC26)

Introduction to the Ball and Plate Balancing System

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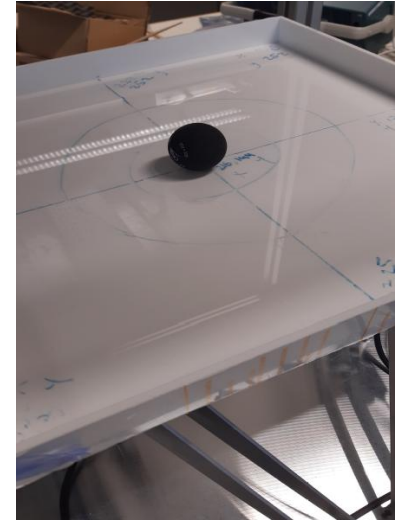
Project work on Ball & Plate (recap kickoff presentation)

Project work as a team:

- No competition intended among teams
- Meetings with staff on technical matters on demand. (By email)
- No weekly project meetings with staff
- Teams organize tasks, objectives and distribution of work themselves.
- Reservations setup through clustermarket

Prior knowledge and assumptions:

- Prior knowledge in core, specialization, elective courses in 3 quartiles of the S&C program



Project work on Ball & Plate (recap kickoff presentation)

Challenge-based learning:

- Define multiple and multi-level ambitions/challenges within your team → Ideas later
- Get organized.
 - Divide tasks and responsibilities
 - Discuss and evaluate team collaboration and responsibilities
 - Make a planning, consider risks and adjust plans when necessary
 - Address organization, technical questions, distribution of work, reporting, presenting, progress, obstacles, etc.
 - → Example project plan later

Objectives

Simple general objective: get things under control . . . (Hint: Make use of your prior knowledge)

Specific learning objectives:

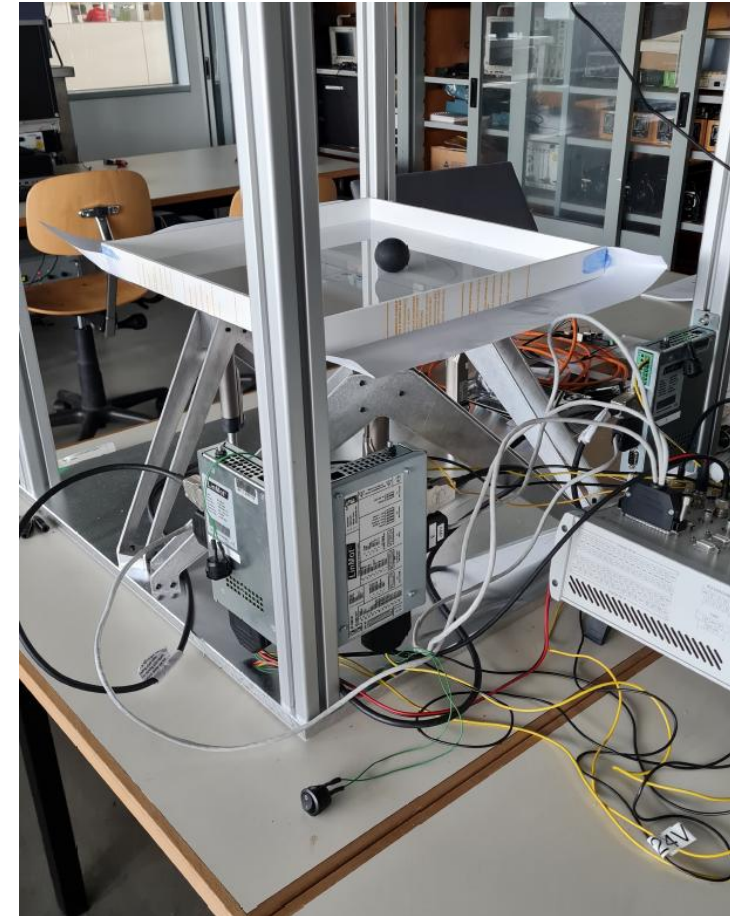
- Modeling
First principles, SID, parameter estimation, experiment design, operational constraints
- Model validation
Experiment design, simulation, quantify accuracy, verify model properties (stability, observability, controllability, linearity, stabilizability, positivity, dissipativity)
- Model-based control design
Choose your own performance specs!! controller synthesis, optimization techniques
- Implementation
Wise choices on analogue/digital conversion, sampling frequency, programming, physical constraints, real-time computational load
- Performance analysis and evaluation
Open and closed-loop analysis, robustness, uncertainty, certification and quantification of performance. demonstration of achieved performance, quantification of performance limitations.

Ball and Plate Balancing system

- 6 DOF system
- Multirate system
- Position control of plate through 3 independent linear motors with encoders
- 2D vision script for detecting ball position

Possible challenge directions include

- Tracking and stability of ball with multiple controllers
- (NEW) 3D camera vision using depth
- (NEW) Vision system design is possible
- (NEW) Multiple ball tracking?
- (NEW) Bouncing ball?
- (NEW) Different balls (colour, weight, size, material)
- What ideas do you have to take this further?



Documentation in Canvas

Start-up documentation

- Equipment manuals will be on canvas **next week**
- Reference material:
https://www.researchgate.net/publication/344687938_Balancing_ball_on_a_plate-Final_Report
- Initial software covers 2D vision and linear motor interfacing for sensors and actuators.
- There is technical assistance, but you need to **ask** for it.

Example project plan

Choose your challenge!

Wk	Objective	Student 1	Student 2	Student 3	Student 4
	Modeling	First principles modeling System identification		Experiment design Data driven identification Operational constraints	
	Model validation	Experiment design, simulation, quantify accuracy, verify model properties			
	Control design	Choose the performance specifications together!!			
		Controller 1	Controller 2	Controller 3 3D vision	Controller 4 3D vision
	Implementation	choices on analogue/digital conversion, sampling frequency, programming, physical constraints, real-time computational load, system operation, system start-up, safety, make readable code All students must be able to operate the system!!			
	Performance analysis and evaluation	Open and closed-loop analysis, robustness, uncertainty, certification and quantification of performance. Demonstration of achieved performance, quantification of performance limitations			

Assign tasks to people with the relevant course background.

Report each objective and discuss with the team so everyone knows each other's topics.

Send the project plan in W3 to p.d.toom@tue.nl and s.weiland@tue.nl

Demonstration